Business Statistics

Module 7
Video 7.4
Standard Deviation and Variance

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Agenda

- Deviation
- Variance

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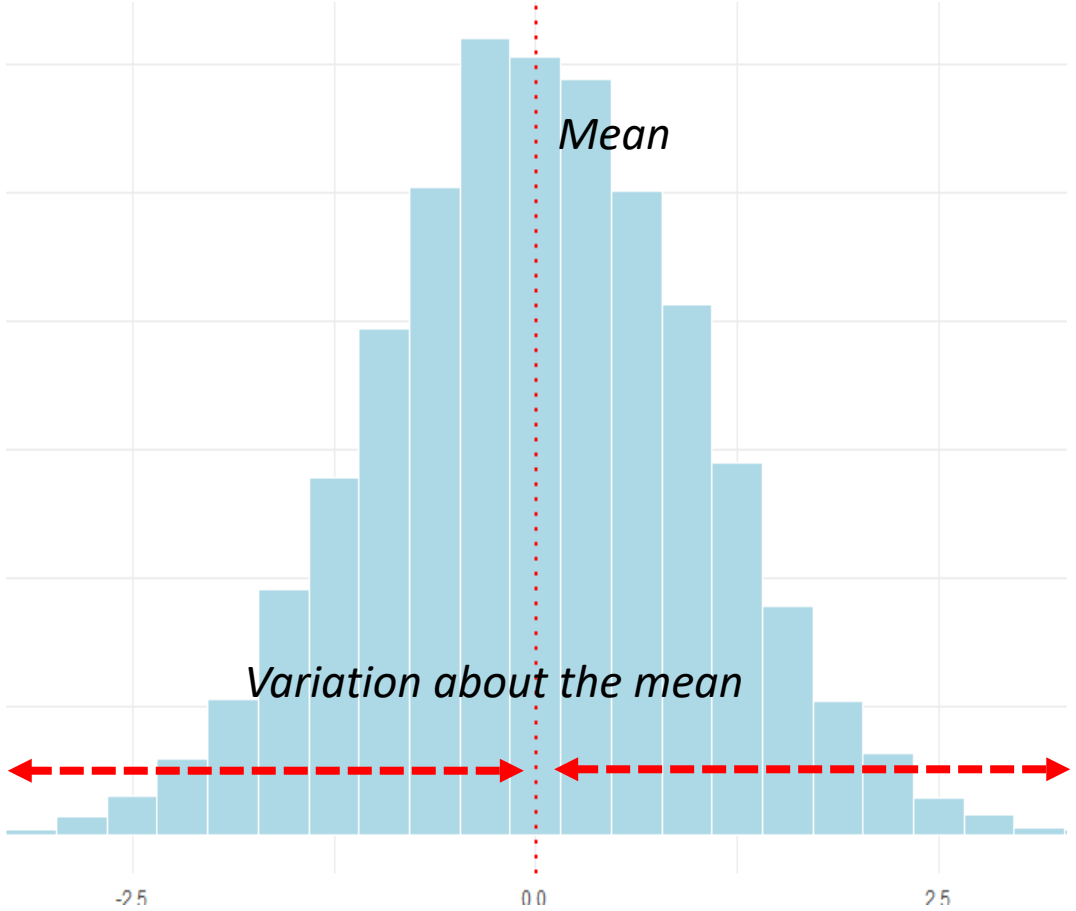
Measures of Variability

- Deviation and Variance are measures of variability in the data
- They are measures of how spread the data is relative to the central tendency of the data



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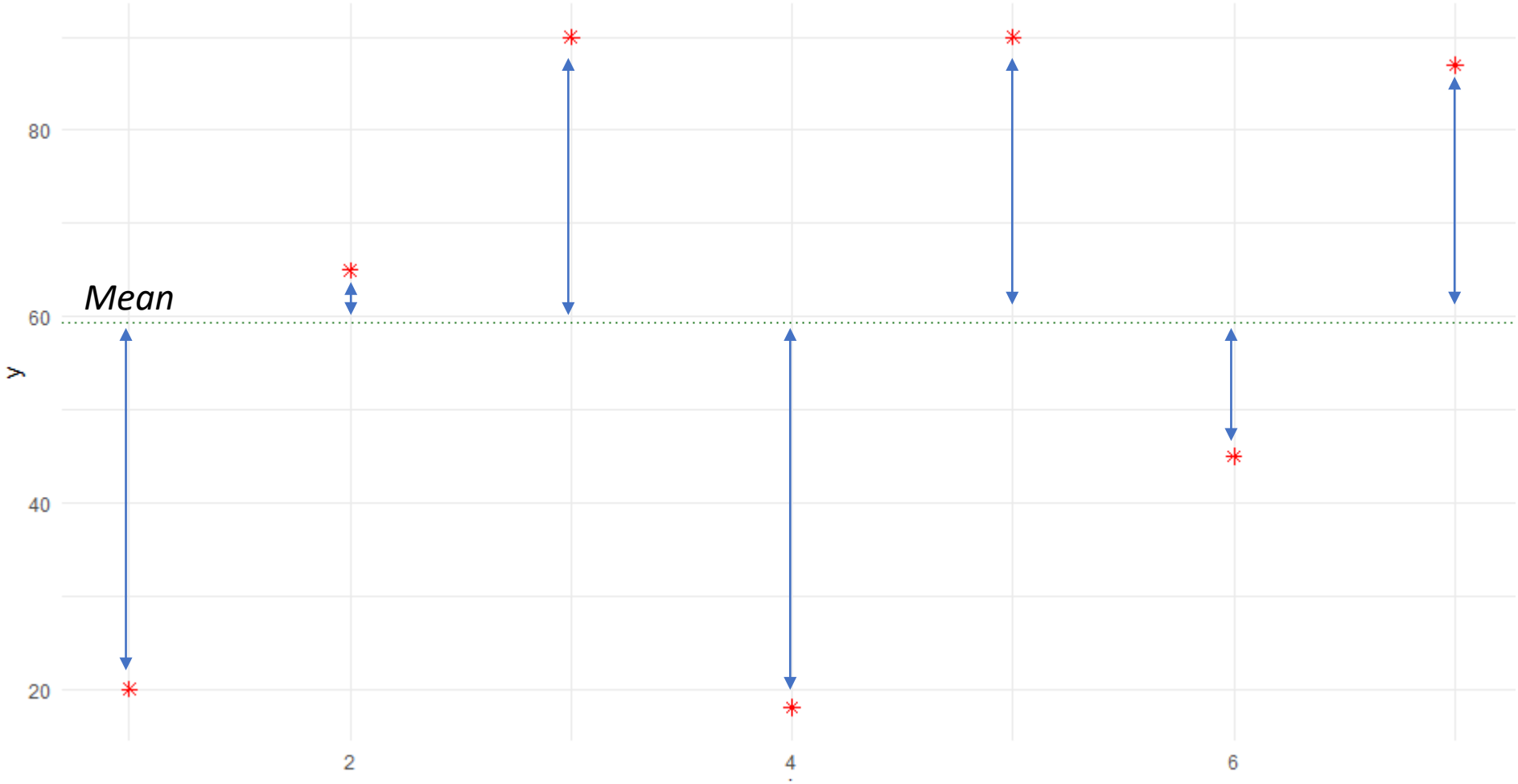
Variability



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Standard Deviation

- Denoted by the Greek letter ***σ*** (Population) and ***s*** (Sample)

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

and

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

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Standard Deviation

Example

y
20
65
90
18
90
45
87

First, we calculate the sample mean

$$\bar{x} = 59.29$$

$$s = \sqrt{\frac{(20 - 59.29)^2 + (65 - 59.29)^2 + \cdots + (87 - 59.29)^2}{6}}$$

$$= 39.99$$

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Mean Deviation

- Mean of the absolute deviations about the mean

$$MD = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$$

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Variance

- Denoted by the Greek letter $\boldsymbol{\sigma^2}$ (Population) and $\boldsymbol{s^2}$ (Sample)

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

and

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

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Coefficient of Variation

- A measure of variability as a ratio

$$CV_{Population} = \frac{\sigma}{\mu}$$

and

$$CV_{Sample} = \frac{s}{\bar{x}}$$

Example: CV= 0.1, The standard deviation is 10% of the mean

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Summary

- Standard Deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

- Mean Deviation

$$MD = \frac{1}{n}\sum_{i=1}^{n}|x_i - \bar{x}|$$

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Summary

- Variance

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

- Coefficient of Variation

$$CV = \frac{\sigma}{\mu}$$

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